

 CAD CROWD VERIFIED SPECIALIST

Mechanical CAD Design & FEA Engineering Portfolio

From initial concept modeling and complex tooling to finite element analysis and production GD&T blueprints.

 Precision Tooling & Molds  FEA & CFD Simulation  Full Manufacturing Drawings

End-to-End Engineering Capabilities



3D CAD & Tooling

Complex progressive dies, plastic injection molds, and production fixtures built for real manufacturability.

SolidWorks • Inventor • Siemens NX



FEA & CFD Analysis

Linear static, gear contact stress, aerodynamic simulation, and structural validation prior to prototyping.

ANSYS • SimScale • FEA Validation

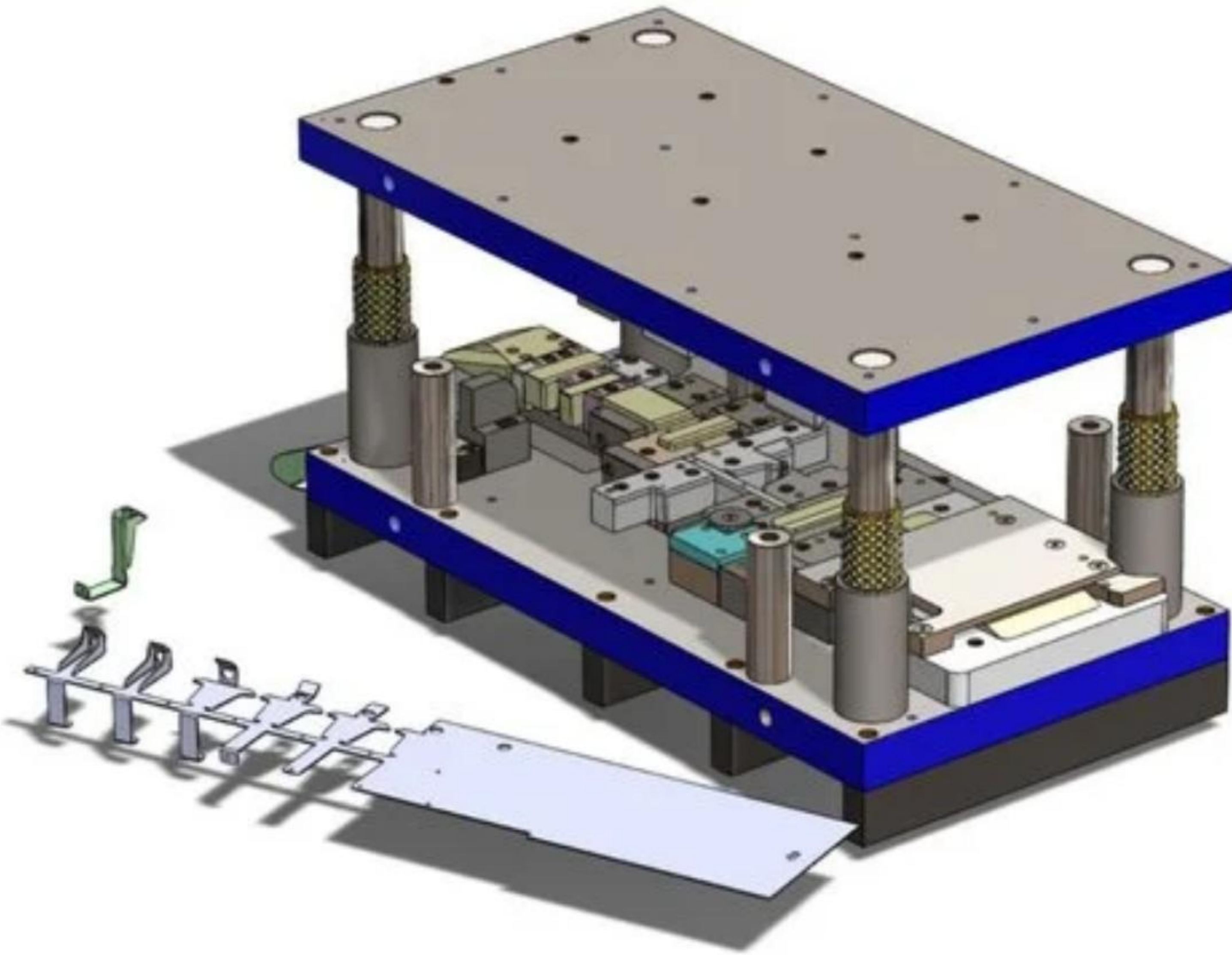


Production Drawings

Detailed 2D fabrication sheets with full GD&T, tolerance stack-up analysis, and machining surface callouts.

ISO / ASME Y14.5 Standards

Progressive Sheet Metal Tool & Die Design



Multi-Station Stamping System

Complete engineering of a progressive stamping die incorporating blanking, piercing, and progressive forming stages.

Optimized Strip Layout

Maximized material yield with scrap reduction

Precision Alignment

Integrated guide pillars and heavy-duty stripper pads

Precision Injection Mold Tooling Design

Core & Cavity Split Architecture

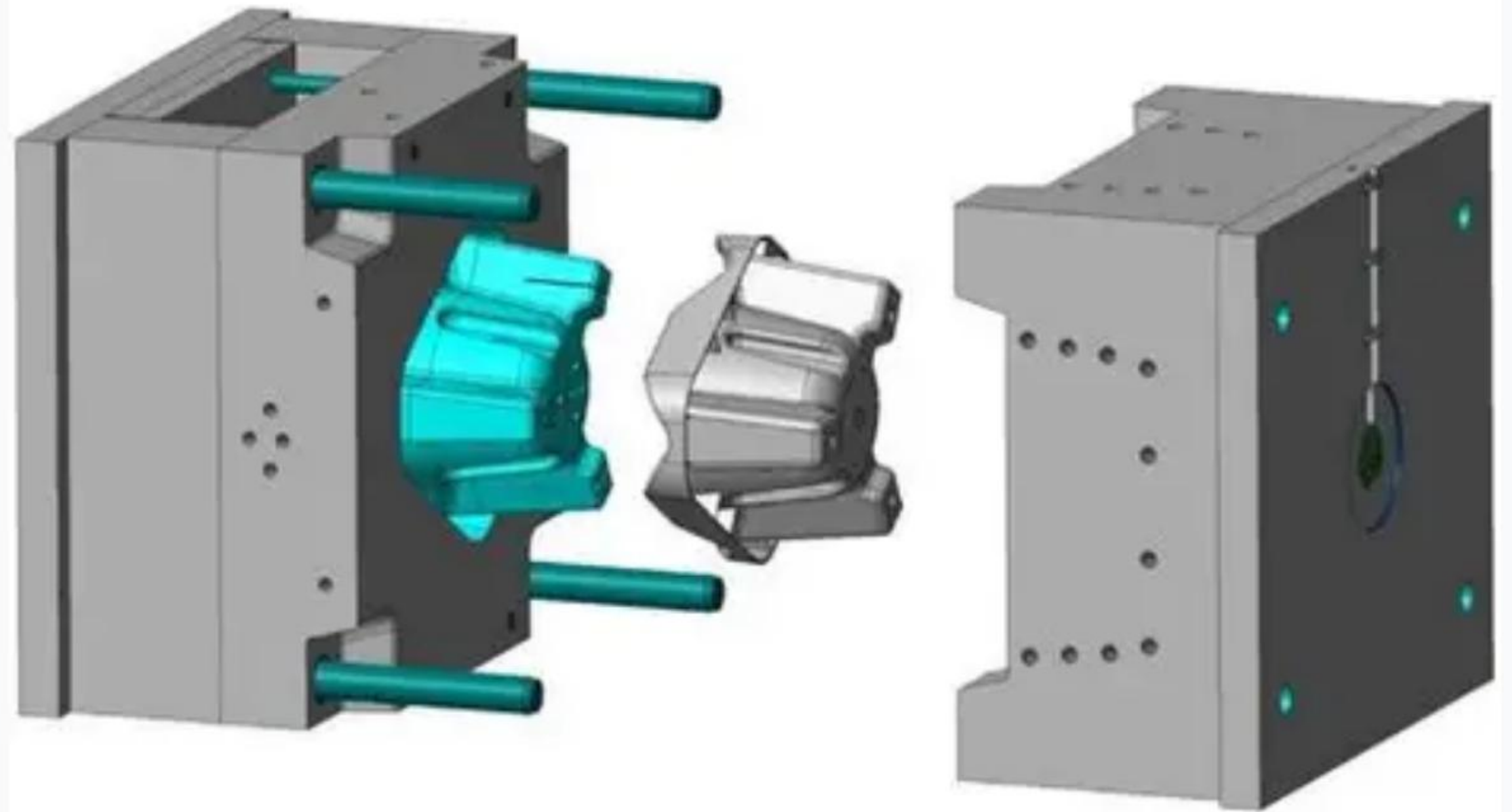
High-precision two-plate injection mold engineered with complex parting lines for consumer product enclosures.

DFM & Draft Validation

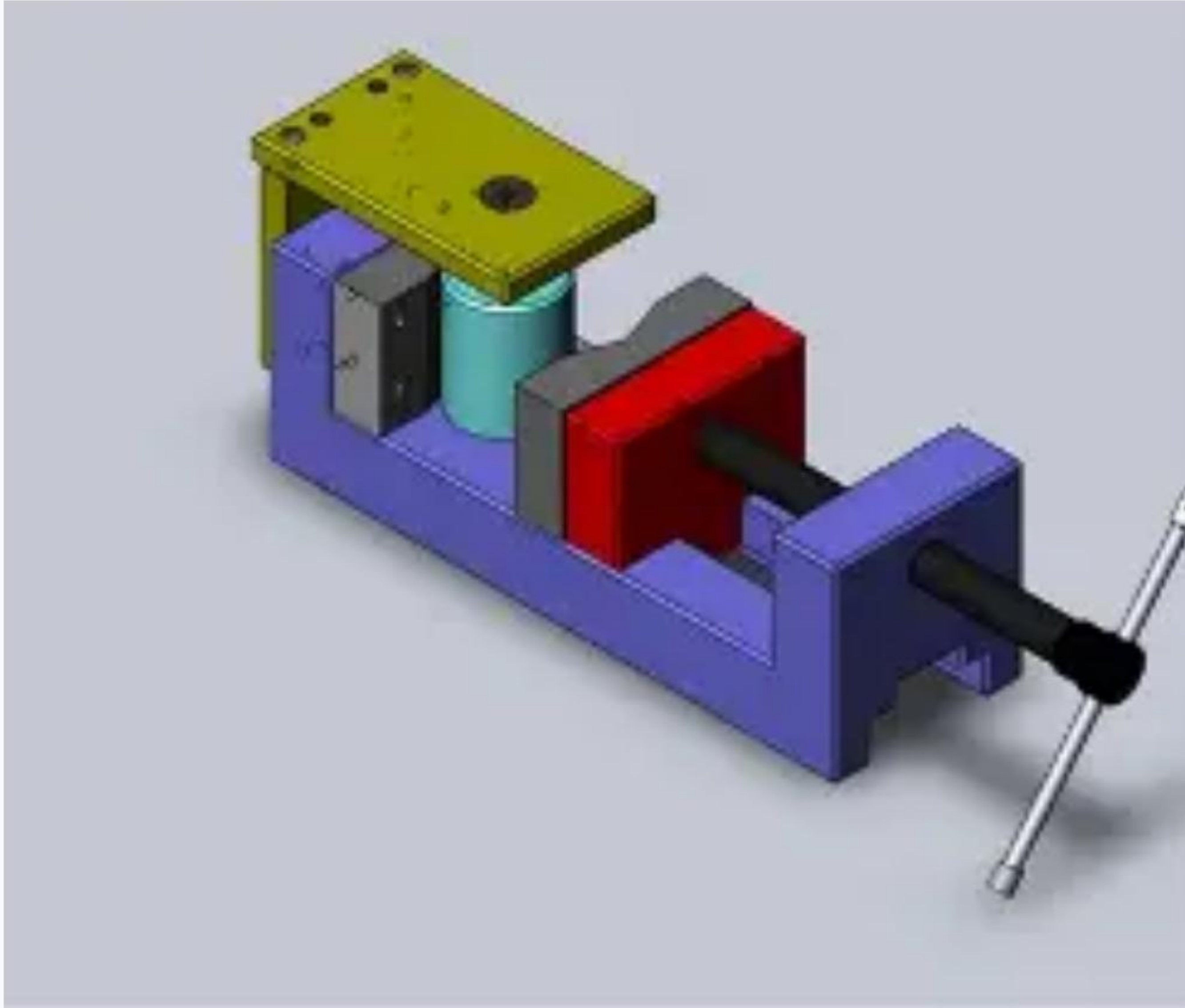
Zero undercuts with verified parting surfaces

Thermal & Ejection Control

Uniform cooling channels and guided pin layout



Custom Machine Vise & Clamping Fixture



Production Tooling & Workholding

Specialized mechanical screw clamp assembly modeled for repeat CNC machining operations with rapid lock engagement.

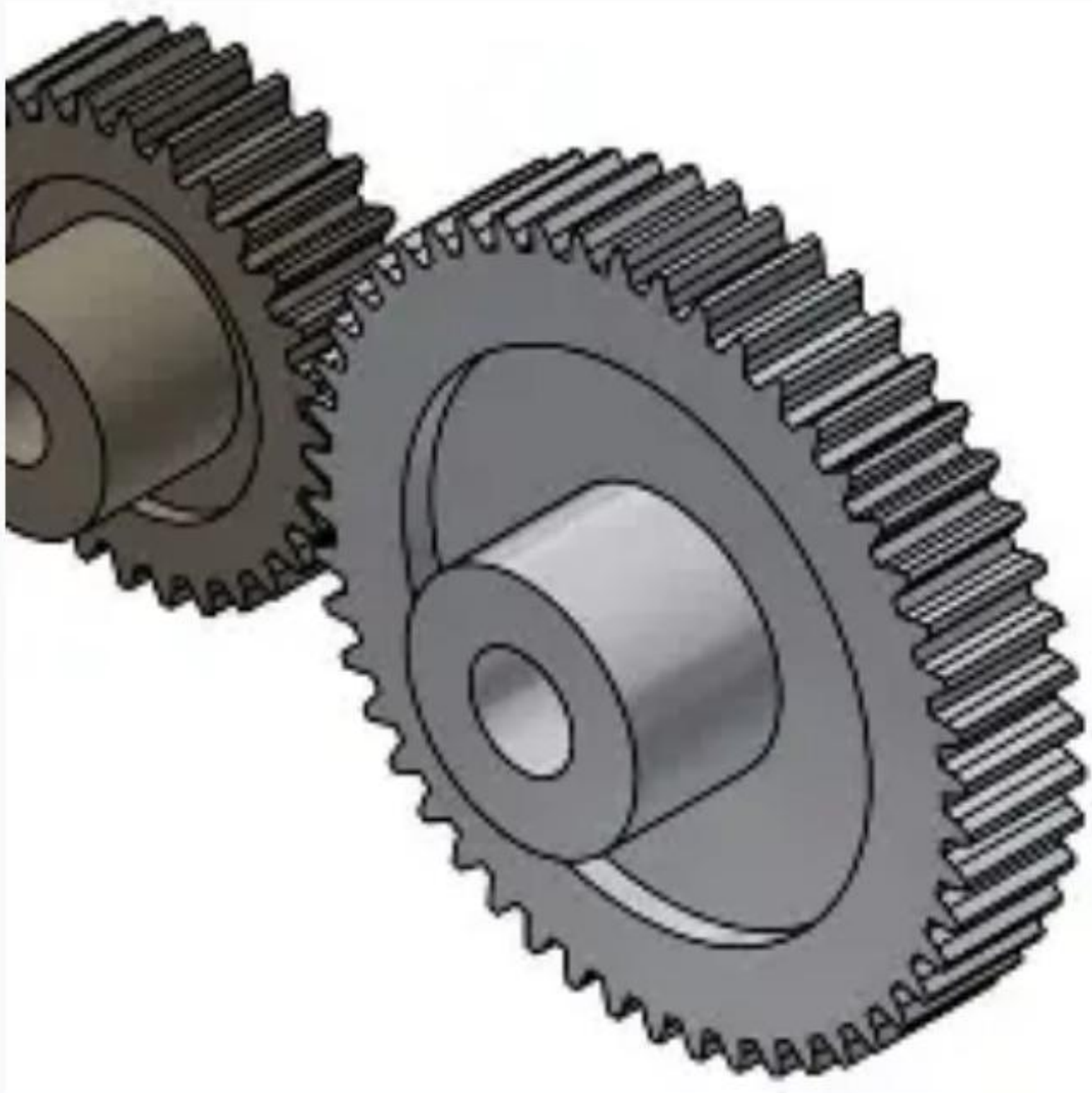
Rigid Clamping Dynamics

Designed to eliminate vibration under heavy cuts

Standard COTS Integration

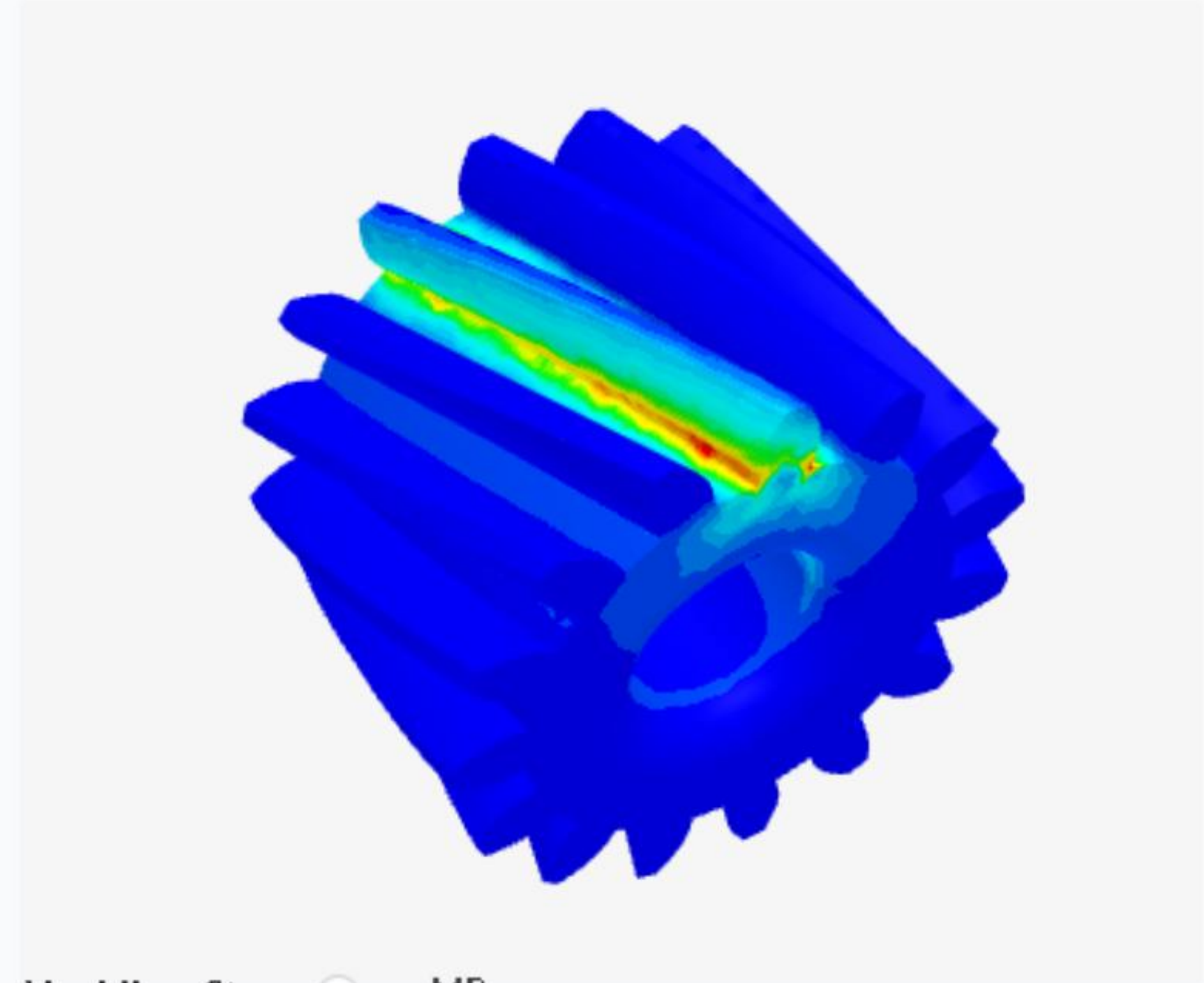
Off-the-shelf lead screws and guide pins for low BOM cost

Power Transmission & Contact Stress Analysis



Parametric Involute Modeling

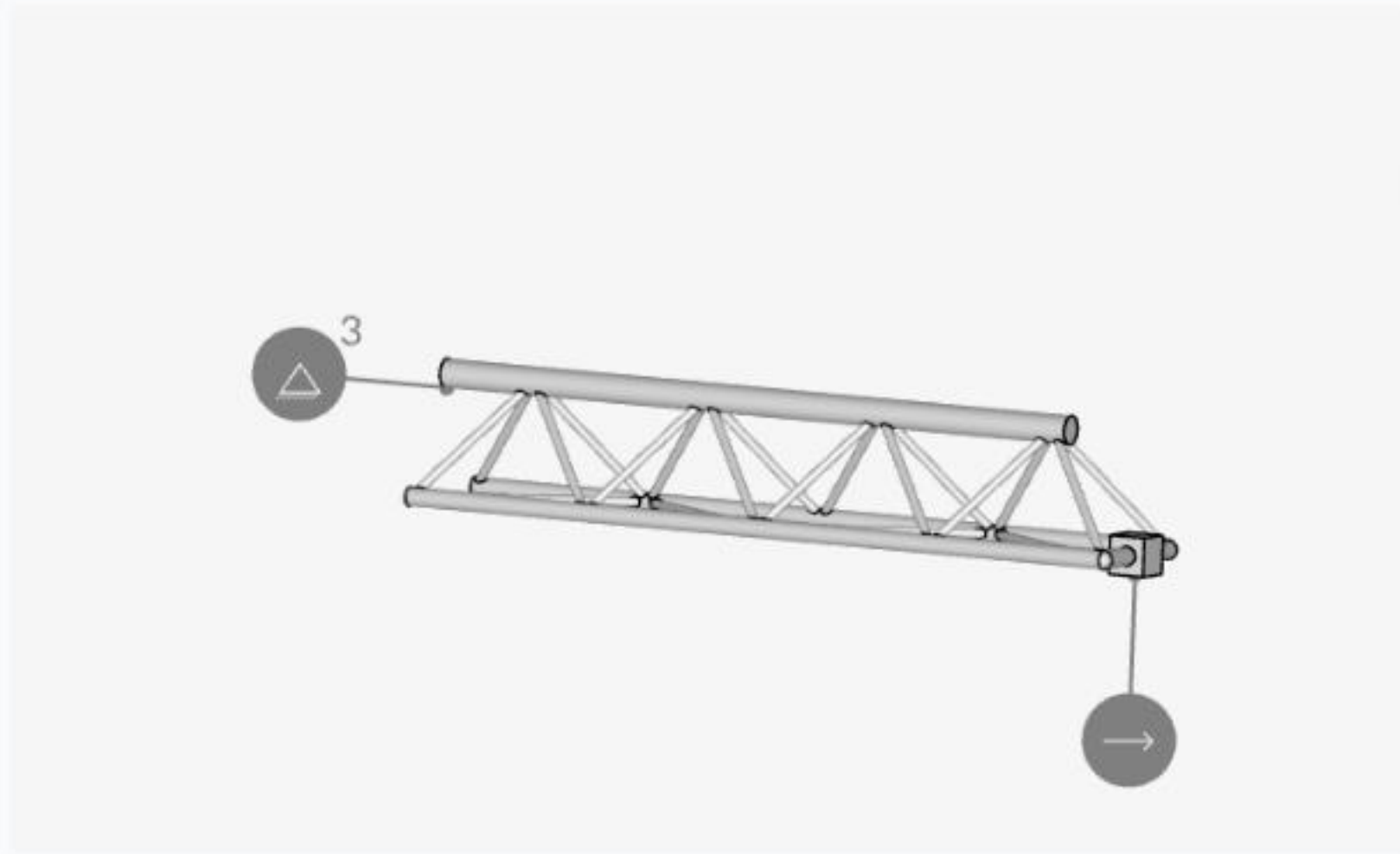
True tooth profile generation and meshing kinematics



Von Mises Stress Verification

Contact stress hotspots identified to prevent pitting & wear

Crane Boom Truss: Linear Static Analysis



Structural Integrity & FOS

Full numerical assessment of a multi-bay tubular crane boom subject to cantilever tip loads and self-weight deflection.

Crane Truss 3D FEA Stress Simulation

Boundary Conditions

Pinned-fixed constraints and concentrated nodal forces

Weight Optimization

Reduced tubular wall thickness without yielding risk

Computational Fluid Dynamics (CFD)

External Flow & Airfoil Simulation

Transonic and subsonic aerodynamic field simulations to evaluate lift, drag profiles, and boundary layer separation.

Pressure Contour Mapping

High-resolution suction and stagnation pressure profiles

Wake & Turbulence Analysis

Boundary layer separation tracking for minimum drag



wing cfd

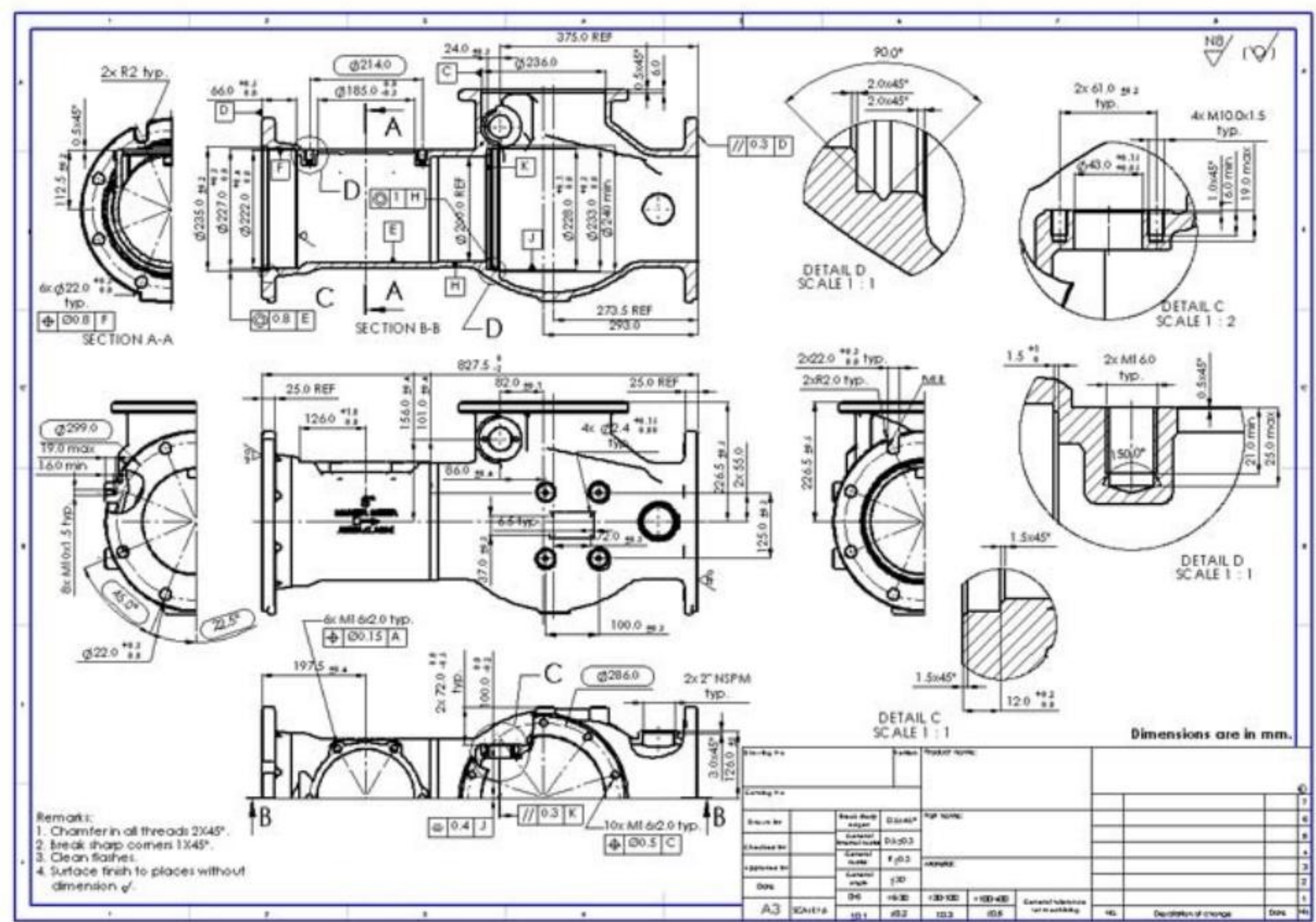
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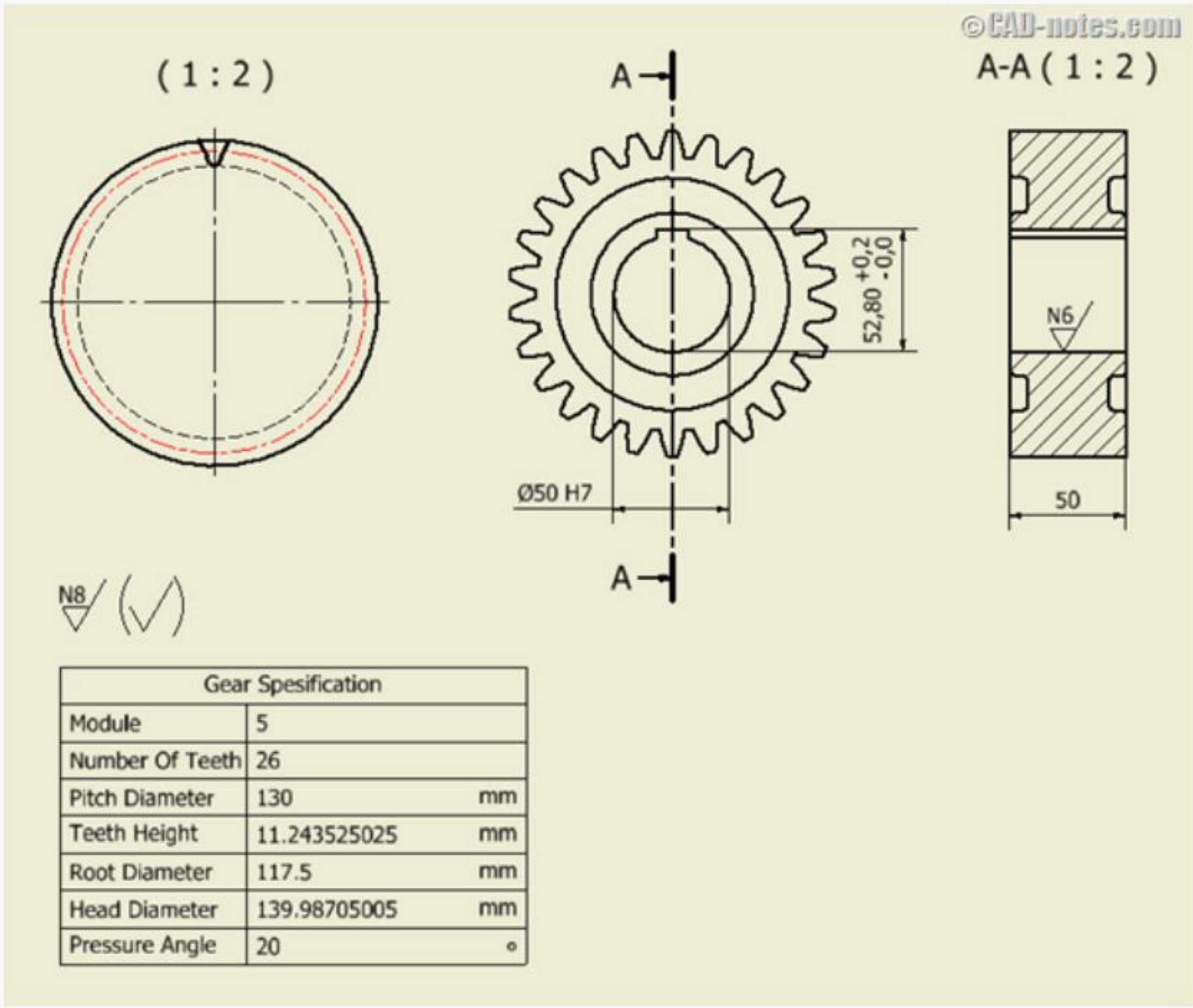
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GD&T & Workshop Manufacturing Drawings



Casing Blueprint with Full GD&T



Machining Tolerances & Specification Tables

Streamlined 4-Step Engineering Delivery

01

Scope & Inputs

Review of client sketches, technical specs, material limits, and manufacturing method.

02

3D CAD & DFM

Parametric solid modeling with draft angle and tooling clearance verification.

03

Simulation Audit

FEA stress and deformation testing to validate structural integrity and eliminate risk.

04

Drawings & Release

Production-ready STEP, IGES, DXF, and PDF drawings handed off with full revision logs.

CLIENT PEACE OF MIND

Why Clients Hire Me on CAD Crowd

100%

On-Time Delivery

Consistent milestone updates and fast turnaround

0

Tooling Clashes

Strict digital mockups and interference checks

$\pm 0.01_{\text{mm}}$

Precision Standards

Tight tolerance compliance for CNC & EDM

24h

Response Time

Clear communication throughout the project lifecycle



Ready to Bring Your CAD Project to Reality?

Available for custom CAD design, tooling engineering, FEA/CFD simulations, and manufacturing drawings on CAD Crowd.



Direct Hire on CAD Crowd



NDA Protected Work



Fast Quote & Delivery